

FIN ST2082–ST2171

Equivariant Second-Differential Classification,
Hodge-Spectral No-Go and the Three-Point Source Frontier

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Six adaptive research rounds — 90 programmes
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This report executes the equivariant-second-differential problem selected in Release 11.06. It computes the exact D_{12} representation-theoretic dimension, imposes locality, integral cellular incidence and refinement intertwining, then audits positivity, Hodge rescaling, spectra and continuum claims. The resulting no-go identifies the missing information type as an intrinsic ternary/2-form source rather than another pairwise-kernel function.

Abstract

ST2082–ST2171 classify possible second differentials $d_1 : C^1 \rightarrow C^2$ for the strict FIN carrier. On the full oriented simplex, exact D_{12} character theory gives

$$\dim \operatorname{Hom}_{D_{12}}(C^1, C^2) = 615.$$

Since the gradient image is isomorphic to $C^0/\mathbf{1}$, the maps satisfying $d_1 d_0 = 0$ form

$$\operatorname{Hom}_{D_{12}}(C^1/\operatorname{im} d_0, C^2)$$

of exact dimension

$$\boxed{517}.$$

Thus equivariance and the chain condition alone are radically nonunique.

Boundary locality collapses the problem. On one oriented triangle, a local row $ax_{ij} + bx_{jk} + cx_{ik}$ annihilates all gradients if and only if it is proportional to the standard oriented boundary $(1, 1, -1)$. D_{12} equivariance leaves one scalar for each of the twelve triangle orbits, producing a 12-dimensional family. Requiring primitive integral cellular incidence restricts every nonzero orbit scalar to ± 1 , and face orientation absorbs the signs. Therefore standard d_1 is combinatorially unique only after a face complex and integral cell basis have already been chosen.

Refinement does not supply the missing choice. The cochain intertwiner fixes the coefficients of lifted triangles but every product-square row annihilates pulled-back base edge cochains. Six vertical square-orbit coefficients remain at level 24 and at least thirteen after level 48. Primitive cellular incidence again yields an associative standard coboundary, but only relative to the chosen product-cell history.

The distinction between differential normalization and Hodge measure is essential. Writing a row scale c_f and positive face weight m_f , the curvature operator depends only on

$$h_f = m_f c_f^2.$$

Row/Hodge inverse rescaling is an exact gauge of description. After quotienting it, physical positive quadratic forms still form an $(\mathbb{R}_+)^{12}$ orbit cone. All positive orbit scales preserve the same cohomology kernel, so topology does not select them.

For the unweighted full simplex an exact identity holds:

$$L_1 = d_0 d_0^* + d_1^* d_1 = 12I_{66}.$$

The standard one-form spectrum is completely flat. Positive orbit Hodge weights split it into 33 distinct rounded eigenvalues in an explicit witness without changing H^1 . Hence no nonzero gauge spectral prediction is invariant over the admissible Hodge cone. Spectral actions and effective dimensions add further cutoff/scale dependence.

The classification can be summarized as

$$517 \xrightarrow{\text{locality}} 12 \xrightarrow{\text{primitive integral cells}} 1,$$

where the final arrow is conditional on exactly the face/cell structure that strict FIN has not sourced. Pairwise kernel and 0-form Dirichlet data contain no independent three-point face observable. The highest-value next theorem is therefore to derive or refute an intrinsic strict ternary cumulant/associator or cycle observable with a D_{12} transformation law, nonzero orbit witness and refinement coherence. Without such a new information type, unique gauge geometry cannot emerge from the current pairwise data without assumption.

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1 Round I: ST2082–ST2096 — exact character classification

Let C^0, C^1, C^2 be oriented vertex, edge and triangle cochain spaces of the full 11-simplex on twelve vertices. Their dimensions are

$$12, \quad 66, \quad 220.$$

The dihedral group acts with orientation signs. For a group element g , the character $\chi_k(g)$ is the signed count of oriented $(k+1)$ -vertex subsets fixed as sets by g . Exact enumeration over all 24 group elements gives the oriented character table used in this report.

The dimension of equivariant maps is the character inner product:

$$\dim \operatorname{Hom}_{D_{12}}(C^1, C^2) = \frac{1}{24} \sum_g \chi_1(g) \chi_2(g) = 615.$$

For the connected complete graph,

$$\operatorname{im} d_0 \cong C^0 / \mathbf{1},$$

with character $\chi_0 - 1$. Therefore the edge quotient has character $\chi_1 - \chi_0 + 1$ and dimension 55. Maschke splitting gives

$$\dim\{T : C^1 \rightarrow C^2 \mid T \text{ equivariant, } Td_0 = 0\} = \frac{1}{24} \sum_g (\chi_1 - \chi_0 + 1) \chi_2 = 517.$$

Theorem 1 (equivariant chain-map no-go). *D_{12} equivariance and $d_1 d_0 = 0$ leave a 517-dimensional real solution space on the full triangle representation. They cannot determine a unique second differential.*

Reflection orientation signs are essential; an unoriented fixed-set census would compute a different and inappropriate representation.

2 Round II: ST2097–ST2111 — boundary locality

For one oriented triangle $i < j < k$, take

$$(Tx)_{ijk} = ax_{ij} + bx_{jk} + cx_{ik}.$$

On a gradient $x_{ij} = f_j - f_i$, exact symbolic coefficient matching gives

$$(a, b, c) = t(1, 1, -1).$$

Thus a boundary-local row satisfying $Td_0 = 0$ is one-dimensional. Equivariance makes t constant on each of the twelve triangle orbits, so locality reduces the 517-dimensional space to twelve dimensions.

Setting all twelve scales to one gives the standard simplicial coboundary. If every present face row must be a primitive integral cellular boundary, its scale is ± 1 . Reversing the face orientation absorbs the signing. Hence:

Theorem 2 (conditional cellular uniqueness). *For a fixed oriented face complex with primitive integral incidence, the boundary-local second differential is unique up to face orientation.*

This does not select the face complex. Full S_{12} symmetry would make all triangles one orbit, but it does not preserve the weighted strict operator. Functions of strict edge weights can assign twelve distinct natural scales.

Nonzero row rescaling leaves $\ker d_1$ unchanged, so H^1 cannot select row normalizations.

3 Round III: ST2112–ST2126 — refinement intertwiners

Let R_1 pull a base edge cochain to identical horizontal copies and set vertical edges to zero. Let R_2 duplicate base face cochains and set product squares to zero. The refinement condition is

$$R_2 d_{1,12} = d_{1,24} R_1.$$

It fixes each lifted triangle coefficient to the corresponding base-orbit coefficient. Every square boundary evaluates to zero on $R_1 C^1$, so square row coefficients are absent from the equation. Six square-orbit coefficients remain at level 24.

At the next binary refinement, all old coefficients are inherited while at least seven new vertical edge-type coefficients appear. The two-level vertical moduli space has dimension at least 13 without primitive integrality.

If a product cell complex is chosen and primitive integral incidence imposed, all rows become standard ± 1 cellular boundaries. The resulting coboundary is associative and satisfies $d_1 d_0 = 0$ at every level. This is a conditional positive construction relative to the cell/refinement history, not a strict selection.

4 Round IV: ST2127–ST2141 — Hodge and positivity

There are two equivalent first-order conventions:

$$\begin{aligned} A &= d_0^* M_1 d_0, & M_1 &= \text{diag}(w_e), \\ A &= (R d_0)^* (R d_0), & R &= \text{diag}(\sqrt{w_e}). \end{aligned}$$

In the weighted-differential convention the compatible second differential is

$$d_{1,w} = d_1 R^{-1},$$

so $d_{1,w} R d_0 = 0$. Weighting d_0 without compensating d_1 would violate the chain condition.

For a face metric $M_2 > 0$, the curvature quadratic operator is

$$Q_1 = d_1^* M_2 d_1 \geq 0.$$

If face row f is scaled by c_f and its metric weight by m_f , the operator depends only on

$$h_f = m_f c_f^2.$$

The transformation

$$c_f \mapsto a_f c_f, \quad m_f \mapsto a_f^{-2} m_f$$

is an exact descriptive redundancy. Quotienting it leaves one positive quadratic coefficient per triangle orbit:

$$h \in (\mathbb{R}_+)^{12}.$$

Every strictly positive orbit choice has the same d_1 kernel of dimension 11 on edge cochains. Topology fixes zero modes but not positive eigenvalues.

5 Round V: ST2142–ST2156 — spectral no-go

For the unweighted complete simplex, exact matrix multiplication gives

$$d_0 d_0^* + d_1^* d_1 = 12 I_{66}.$$

All one-form Hodge modes have eigenvalue 12. This fully symmetric choice has no nontrivial dispersion structure.

Assigning different positive scales to the twelve D_{12} triangle orbits preserves $H^1 = 0$ but splits the one-form spectrum. An explicit deterministic choice produces 33 distinct rounded eigenvalues ranging from approximately 12 to 43.21. Continuous changes of the Hodge cone continuously tune these values.

Consequently:

- cohomology controls only zero-mode dimensions after a complex is chosen;
- positive gauge stiffness/mass scales remain free;
- $\text{Tr } f(L_1/\Lambda^2)$ depends on twelve Hodge parameters, the cutoff function and scale;
- one-form heat traces and effective dimensions are measure dependent;
- refinement flag zero modes can be removed with product squares only by introducing new vertical spectral parameters;
- no Lorentz signature, wavevector locality or Maxwell normalization is produced.

Theorem 3 (Hodge-spectral prediction no-go). *No nonzero one-form spectral prediction is invariant over the currently admissible positive D_{12} -natural Hodge cone. Integral incidence and fixed cohomology do not select gauge dynamics.*

6 Round VI: ST2157–ST2171 — final frontier

The exact reduction ladder is

$$517 \xrightarrow{\text{boundary locality}} 12 \xrightarrow{\text{primitive integral cells}} 1.$$

The final one-dimensional result is conditional on a chosen cell complex and integral cellular basis. Refinement adds unconstrained vertical coefficients, and Hodge quotienting leaves twelve positive physical quadratic parameters at the base level.

Object	Status	Boundary
equivariant chain maps	[Classified]	dimension 517
boundary-local maps	[Classified]	dimension 12
primitive cellular d_1	[Conditional unique]	chosen cells
refinement intertwining	[Nonunique]	vertical dimensions 6, then ≥ 13
positive Hodge operator	[Constructed family]	$(\mathbb{R}_+)^{12}$
standard full-simplex spectrum	[Proven flat]	$12I_{66}$
weighted gauge spectrum	[Tunable]	no strict prediction
continuum gauge physics	[Not established]	scale/locality/OA absent

7 Most important result

The second differential itself is not the missing unique object once cellular integrality is imposed; it becomes the standard boundary operator on whatever cells were already chosen. The missing information is the selection and physical weighting of ternary/2-form structure.

Pairwise kernel data and the 0-form Dirichlet operator contain no independent three-point observable. Any triangle weight built afterward as a function of three pair weights is another chosen rule, and Release 11.05 exhibited many nonproportional natural rules.

Thus the decisive obstruction is now typed as:

$$\text{strict FIN lacks an intrinsic ternary/2-form source.}$$

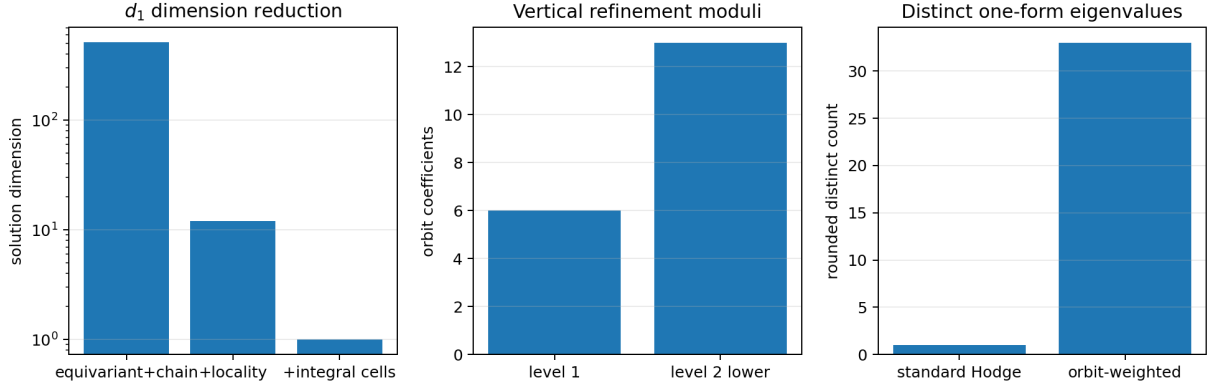


Figure 1: Left: exact dimension reduction under progressively stronger conditions. Centre: vertical moduli introduced by refinement. Right: the flat standard Hodge spectrum versus a natural orbit-weighted witness.

8 Highest-value next theorem

The new source gate requires:

- T1. an intrinsic strict ternary/cycle observable;
- T2. a D_{12} transformation law and nonzero orbit witness;
- T3. face-set and d_1 selection independent of target fitting;
- T4. a positive Hodge measure source;
- T5. $12 \rightarrow 24 \rightarrow 48$ refinement transport/coherence;
- T6. dimensional/continuum Maxwell scaling;
- T7. an OA platform and record.

The current strict repository passes zero complete rows. Existing chiral bispectra are valuable orientation markers but remain premise-based and do not supply the complete face/Hodge/refinement package.

The highest-value next theorem is therefore:

derive or refute an intrinsic ternary cumulant, associator or cycle response from the adaptive/refinement law, then test whether its orbit tensor selects a face complex and positive Hodge measure without target fitting.

A nonzero, refinement-coherent source would introduce genuinely new information beyond the pair kernel and could change the physical verdict. Exhausting the available strict ternary candidates with no passing source would be a total no-new-live-frontier result for the gauge-geometry route.

9 Route ranking

Route	Rigorous-success estimate	Physics decision power
strict ternary/2-form source intake	0.55	1.00
conditional integral product-cell gauge model	0.90	0.50
fine-holonomy measurement of Hodge weights	0.40	0.60

Route	Rigorous-success estimate	Physics decision power
additional pairwise face functions	0.05	0.20
OA frozen-channel experiment	0.45	0.55

These are strategic judgments, not empirical probabilities.

10 Final interpretation

For a chosen cellular complex, FIN admits a canonical primitive second differential and coherent conditional Hodge/gauge models. Neither equivariance, locality, refinement, positivity, cohomology nor spectra choose the cells and their Hodge weights from current pairwise data. FIN remains a finite Dirichlet–Markov–spectral information geometry with conditional gauge extensions, not a fundamental physical theory. The intrinsic ternary-source audit is the next point capable of changing that conclusion.

A Six-round ledger

Range	Target	Verdict
ST2082–ST2096	equivariant d_1	exact dimension 517
ST2097–ST2111	boundary locality	twelve orbit scales
ST2112–ST2126	refinement	vertical coefficients persist
ST2127–ST2141	positivity/Hodge	rescaling gauge and positive cone
ST2142–ST2156	spectra/cohomology	flat/tunable spectrum no-go
ST2157–ST2171	synthesis	intrinsic ternary-source frontier

B Reproducibility

The authoritative result sets run from `FIN_ST2082_ST2096_Results.json` through `FIN_ST2157_ST2171_Results.json`. Each contains fifteen programmes and packet SHA-256 references. The combined test `test_fin_st2082_st2171.py` verifies all ninety packets, character dimensions, locality reduction, refinement moduli, Hodge identity, spectra, updated gate, figure and nonpromotion boundaries.

C Nonclaims

This report does not derive an intrinsic ternary source, a unique face complex, Hodge measure, gauge coupling, continuum Maxwell theory, physical scale, state ontology, platform or evidence. It does not complete legacy role transfer, QW-2191, Standard Model, gravity, L_{total} or Theory of Everything. All future FIN PDFs remain English.

References

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